

ECONOMICS & MANAGEMENT **(HS-412)**

ASSIGNMENT # 01

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Section: A

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Semester: 7th Semester

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Question 01:

Relate linkage between **Money - Time relationship**. Identify reasons that lead to devaluation with passage of time.

Answer:

1. Linkage Between Money and Time:

The relationship between **money and time** is described by the **Time Value of Money (TVM)**.

It means that **a dollar today is worth more than a dollar in the future**, because today's money can be invested to earn returns.

Reasons for the Time Value of Money:

- **Investment Opportunity:** Money available now can be invested in business or savings to earn profit or interest.
- **Future Uncertainty:** The future is uncertain due to inflation, risk, and opportunity cost — meaning future money may buy less.
- **Compounding Effect:** Returns earned on money can be reinvested to generate more returns over time, increasing total value.

Mathematical Relationship:

$$FV = PV(1 + r)^n \text{ and } PV = \frac{FV}{(1 + r)^n}$$

Where:

- **FV** = Future Value
- **PV** = Present Value
- **r** = Interest rate
- **n** = Time period

In short, money grows with time if it earns returns, but loses value if it remains idle.

2. Reasons for Devaluation of Money Over Time:

Over time, the **purchasing power of money decreases**, meaning it buys fewer goods and services than before.

This happens mainly due to the following reasons:

1. Inflation:

A continuous rise in the general price level is the main cause of devaluation. When prices go up, the same amount of money buys fewer goods and services.

2. Negative Real Interest Rates:

If the inflation rate is higher than the interest earned on savings, the real value of money falls.

For example, earning 2% interest during 4% inflation results in a real loss of 2%.

3. Currency Depreciation:

A decline in the exchange rate of the domestic currency increases the cost of imports and raises domestic prices.

4. Excess Money Supply:

When the central bank or government prints too much money, it reduces the value of each dollar in circulation, leading to inflation.

5. Loss of Confidence and Policy Issues:

If people lose trust in a country's monetary or economic policies, they tend to spend money quickly, increasing inflation and reducing value.

Example:

At 3% annual inflation, **\$100 today** will be worth only **\$74.40** after **10 years**, showing a **25.6% loss** in purchasing power.

Conclusion:-

The **money–time relationship** shows that money's value changes with the passage of time.

Because of factors such as inflation, negative real interest rates, and currency depreciation, money **devalues over time**.

To maintain or increase its real worth, money should be **invested productively** to earn returns that at least match the inflation rate.